



Real-Time Traffic Analytics Using Azure Stream Analytics

Project Information

Project Title: Real-Time Traffic Analytics Using Azure Stream Analytics Course: AI & Data Science Microsoft Data Engineer – DEBI

Round: 3

Group Members & Roles

- **Abdalla Nasser Bekhit Abdalla Ewida:** Team Leader , Technical Team Lead
- **Nour Hassan Hamdy Gamil:** Data Simulation & Python Integration
- **Zahran Alaa Sayed Mohammed:** System Testing & Quality Assurance
- **Rawan Khaled Mohamed Mustafa:** Data Engineering & SQL Queries
- **Sohaila Ahmed Ali Elgmal:** Azure Cloud & Pipeline Architect
- **Menna Muhammad Ali Eissa:** Visualization Lead - Power BI

Project Description

This project aims to build an end-to-end system for the real-time analysis of traffic in Greater Cairo. The system will be based on a realistic data source generated by a custom Python script, which creates data that simulates traffic flow, including real street names, various vehicle types, and realistic speed patterns.

This streaming data will be sent to Microsoft Azure for processing using Azure Stream Analytics. The resulting insights and processed data will then be stored in a cloud database. Finally, key performance indicators (such as congestion zones, average speeds, and alerts) will be displayed on an interactive dashboard using Power BI.

Objectives

- **Activate and utilize** the script to generate realistic, streaming traffic data.
- **Build a data pipeline** to receive data from the script and ingest it into Azure Event Hubs.
- **Develop queries** in Azure Stream Analytics to analyze the data in real-time and detect important patterns (e.g., congestion, accidents).
- **Store** the processed data and results in an Azure SQL Database.
- **Design and develop** an interactive dashboard in Power BI to display live metrics and alerts.

Tools & Technologies

- **Data Source:** Python Script.
- **Data Ingestion:** Azure Event Hubs.
- **Real-Time Processing:** Azure Stream Analytics.
- **Storage:** Azure SQL Database.
- **Visualization:** Power BI.

Stakeholder Analysis

This project involves several stakeholders who play important roles in its development and potential use.

The project team members are responsible for designing, developing, and testing the full data pipeline — from data simulation using Python to real-time processing on Azure Stream Analytics and visualization in Power BI.

The mentors and supervisors provide technical guidance and ensure the project meets academic and quality standards.

The Ministry of Communications and Information Technology (MCIT), through the Digital Egypt Pioneers Initiative (DEPI), acts as the project sponsor and platform provider, offering the environment and tools for implementation and evaluation.

The traffic authorities represent potential end users who could benefit from the final analytical dashboard to monitor traffic conditions and identify congestion trends in real time.

Finally, the general public benefits indirectly from improved traffic management and decision-making supported by the project's analytical insights.

Database Design

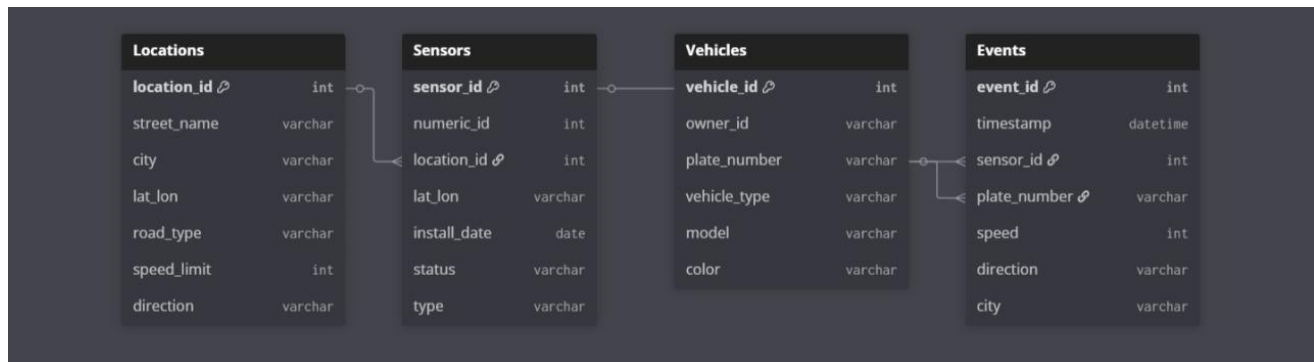
The database consists of four main tables: Locations, Sensors, Vehicles, and Events. These tables are designed to store and manage traffic monitoring data efficiently.

- **Locations Table:**
Stores details about different road segments, including street name, city, geographic coordinates, road type, speed limit, and direction.
- **Sensors Table:**
Contains information about sensors installed at various locations. Each sensor is linked to a specific location through the `location_id` field and includes details such as installation date, type, and operational status.
- **Vehicles Table:**
Holds records of registered vehicles, including their unique plate numbers, owner IDs, vehicle types, models, and colors.
- **Events Table:**
Records traffic events detected by the sensors. Each event includes a timestamp, the related sensor ID, the vehicle's plate number, the recorded speed, direction, and city of occurrence.

Relationships:

- Each location can have multiple sensors (one-to-many).
- Each sensor can generate multiple events (one-to-many).
- Each vehicle can be involved in multiple events (one-to-many).

This relational structure allows efficient tracking of traffic activity, sensor performance, and vehicle movement across various locations.



UI/UX Design

The project's user interface is designed using Power BI to display real-time traffic data processed through Azure Stream Analytics.

The dashboard focuses on showing key metrics and charts that help understand traffic patterns clearly.

It includes:

- A panel showing total events, average speed, and traffic density.
- Charts for speed distribution and vehicle types.
- A time-based graph showing how traffic changes over time.
- A live event table listing the latest traffic data received.

The design is simple, data-focused, and updates automatically as new events are streamed from Azure.